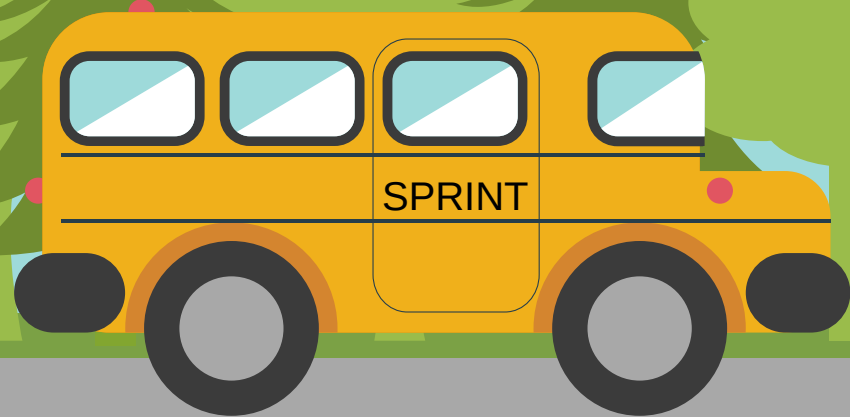


Sprint Community Transport

Final Release Presentation



Who are we?

Beth Cartwright

James Rennie

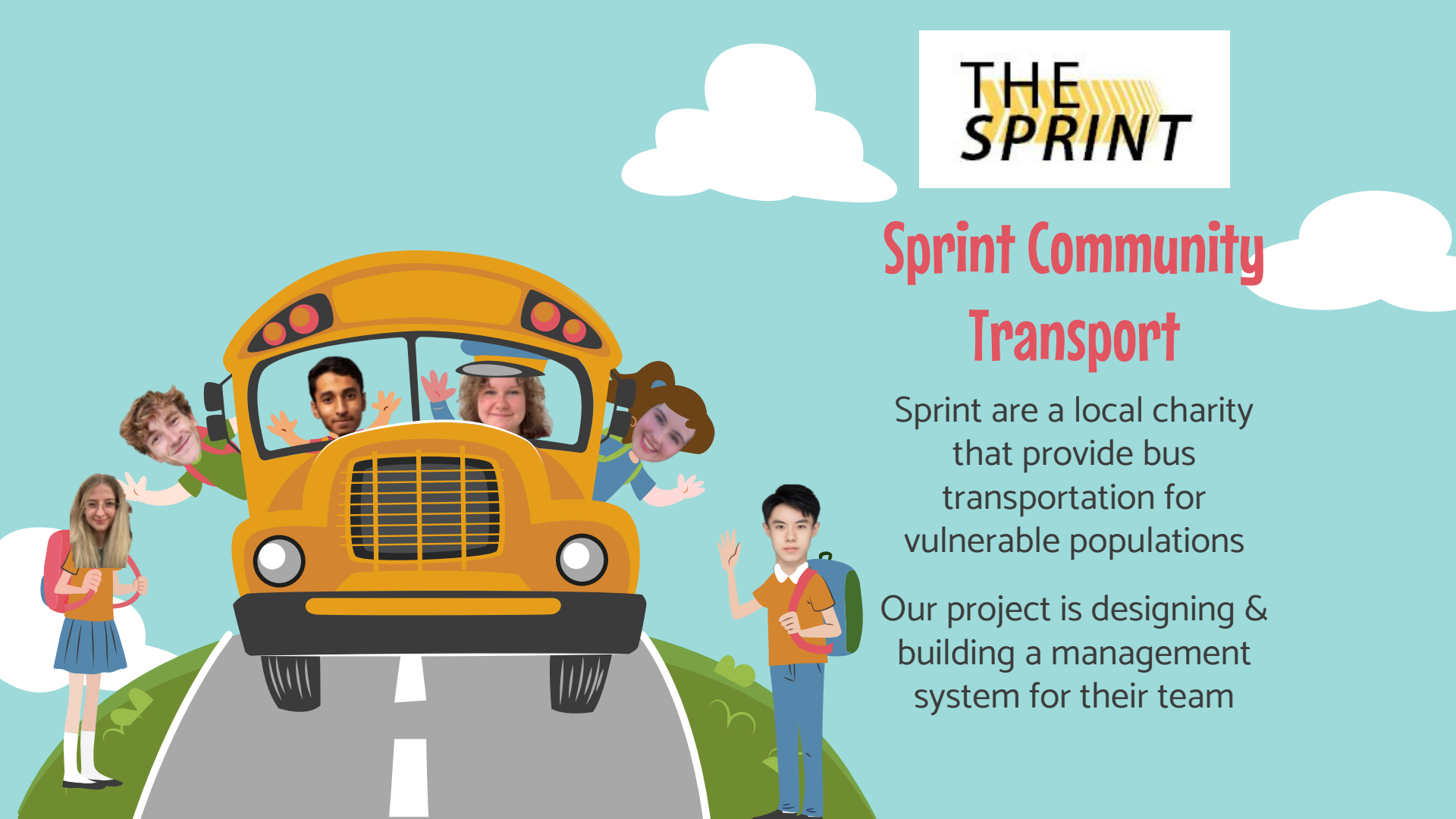
Sajjad Hussain H Alkhamis

Amy Jose

Eva Anderson

Ethan Wang



An illustration of a yellow school bus driving on a grey road with white dashed lines. A male driver and a female co-driver are visible in the front seats, both waving. A young man is leaning out of the back of the bus, also waving. On the left side of the road, a young girl with a red backpack and a blue skirt is walking towards the bus. On the right side, a young man with a blue backpack is walking away from the bus, waving. The background is a light blue sky with two white clouds.

THE SPRINT

Sprint Community Transport

Sprint are a local charity
that provide bus
transportation for
vulnerable populations

Our project is designing &
building a management
system for their team

	A	B	C	D	E	F	G
1	Thursday week 1			date			BOOKINGS WE COULDN'T DO***
2	NAME	ADDRESS	DESTINATION	TIME	PERM?	SPECIAL INSTRUCTIONS	
3	BUS CHECK			8AM			
4	SLA	123 LANE ROAD		8.20		FOLLOW PRINTOUT	
5		123 LANE ROAD					
6	CLA	123 LANE ROAD	SAINSBURYS	10.30	PERM	TROLLEY	
7	DER	123 LANE ROAD			PERM		
8	MCN	123 LANE ROAD			PERM	TROLLEY	
9	PAY	123 LANE ROAD	FULL****		PERM	TROLLEY	
10	JOS	123 LANE ROAD			PERM	TROLLEY	
11	STE	123 LANE ROAD			PERM	TROLLEY	
12	NOR	123 LANE ROAD			PERM	TROLLEY	
13	WED	123 LANE ROAD			PERM	TROLLEY	
14				11.10			
15				11.30			
16				11.45			
17				12.00			
18	TIME TO GET THERE						
19	WED	SAINSBURYS	123 LANE ROAD	12.15	PERM	TROLLEY	
20	MCN		123 LANE ROAD		PERM	TROLLEY	
21	STE		123 LANE ROAD		PERM	TROLLEY	
22	PAY		123 LANE ROAD		PERM	TROLLEY	
23	JOS		123 LANE ROAD		PERM	TROLLEY	
24	NOR		123 LANE ROAD		PERM	TROLLEY	
25	DER		123 LANE ROAD		PERM	TROLLEY	
26	CLA		123 LANE ROAD		PERM	TROLLEY	
27	LUNCH			12.00			
< > MON TUES WEDN THURS FRI MON 2 TUES2 WEDN 2 THURS 2 FRI 2 +							

Currently the schedule is managed through a disorganised excel sheet, where repeated journeys are copied & pasted from week to week.

Current System

(data retracted for GDPR reasons)

User Stories



Office Manager

I want to quickly **add repeating journeys** to a calendar so I can reduce workload when planning routes.

I want to **view upcoming events** so I can check availability and optimize journeys.

I want to **add, remove & edit client details** so I can keep the database accurate and reflect changes in client needs.



Charity Liason

I want to pull **monthly breakdowns** of journey numbers, client ages, and **needs** so I can accurately report to the funding company and show how their support is used.



Driver

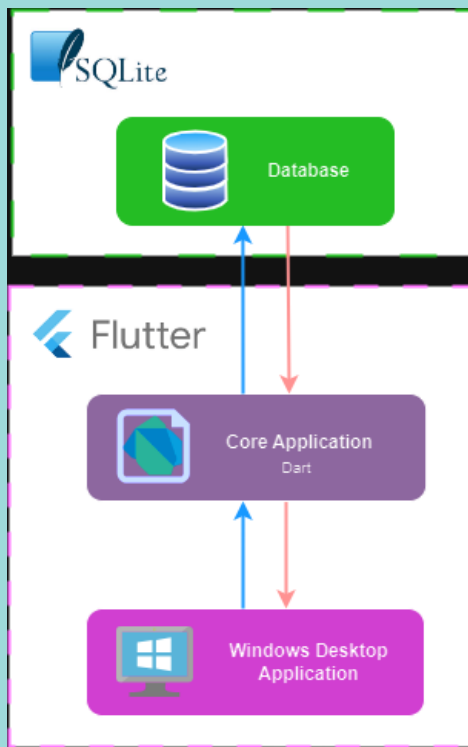
I want to know **which clients** I'm taking **where** and **when** each day so I can plan efficiently and prepare the right information and equipment.

I want to **log** completed and upcoming **vehicle checks** so I can keep vehicles safe, compliant, and schedule maintenance without disrupting journeys.



Demonstration

Tech Architecture



At MVP Release, we were operating with a **locally hosted SQLite database**, integrated with a **flutter** frontend



Actually...

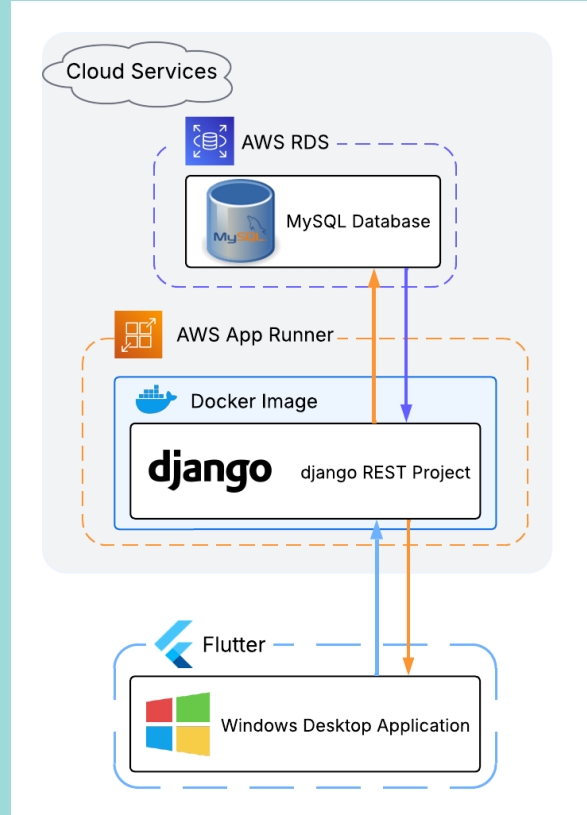
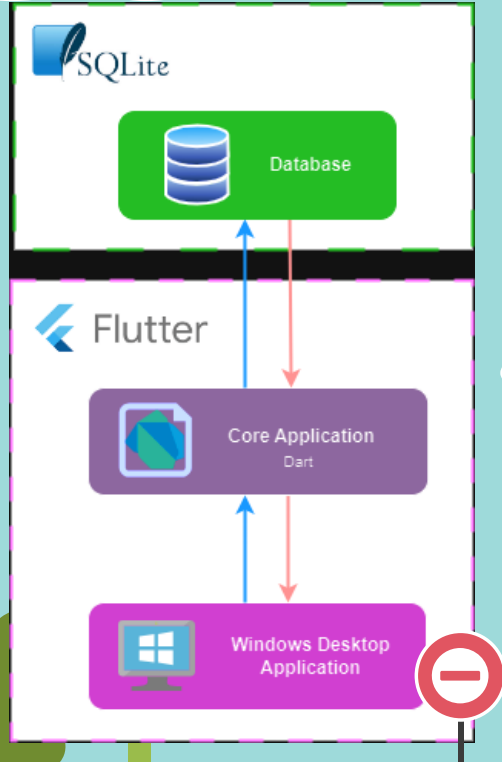


Client

Multiple users will make use of the system!



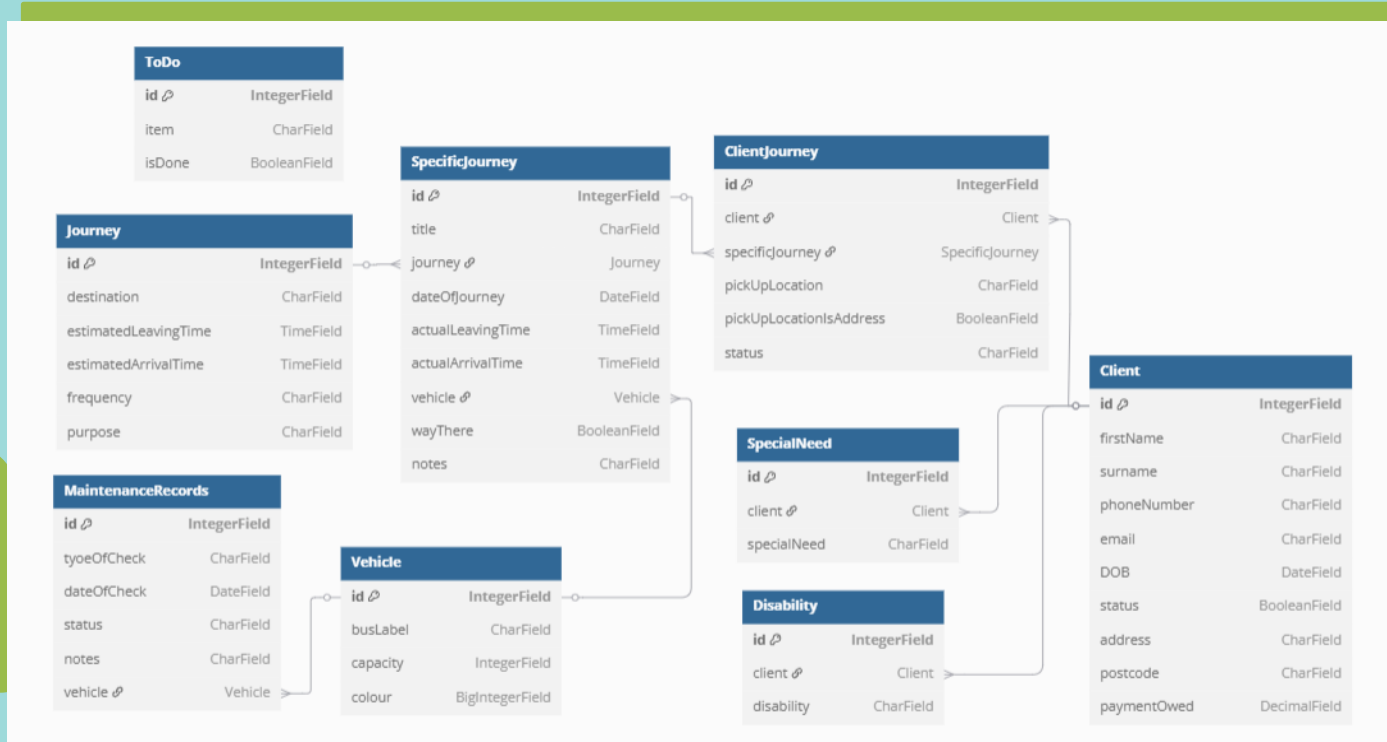
Tech Architecture

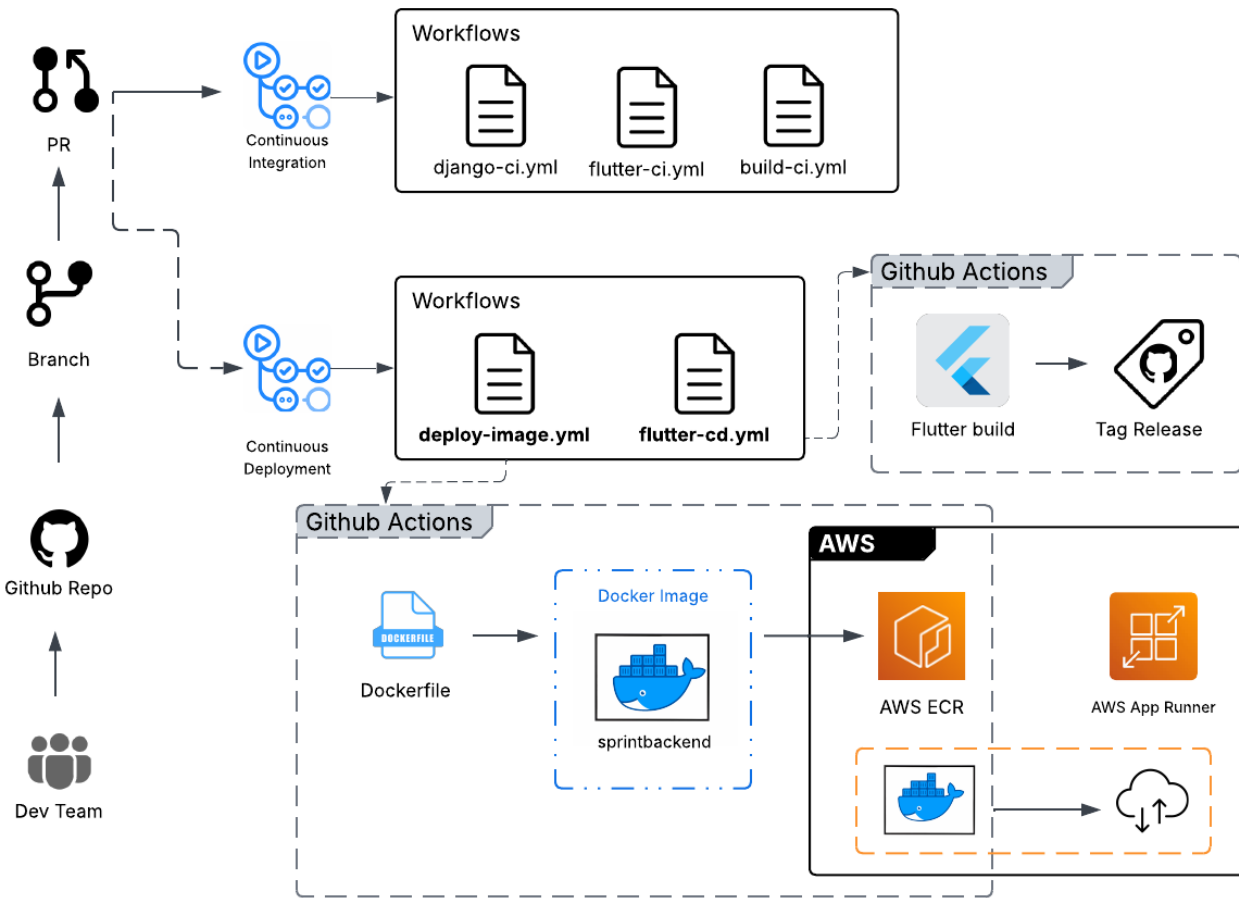


In our Final Implementation, we moved to an **AWS hosted backend** integrated with the **flutter** frontend, to enable multiple users



Entity Relationship Diagram





CI & CD



Development – Test Coverage

LCOV - code coverage report

Current view: [top level - lib](#)

Test: lcov.info

Date: 2025-05-07 16:45:33

	Hit	Total	Coverage
Lines:	1205	2314	52.1 %
Functions:	0	0	-

Filename ↕	Line Coverage	Functions ↕
print_out.dart	0.0 % 0 / 54	- 0 / 0
multiselect_dropdown.dart	0.0 % 0 / 91	- 0 / 0
login_page.dart	1.5 % 1 / 67	- 0 / 0
view_buses.dart	1.6 % 1 / 64	- 0 / 0
main.dart	2.3 % 3 / 132	- 0 / 0
popup_journey_functions.dart	5.2 % 5 / 97	- 0 / 0
popup_general_functions.dart	20.5 % 34 / 166	- 0 / 0
popup_validation.dart	27.5 % 14 / 51	- 0 / 0
popup_inline_multiselect.dart	33.7 % 68 / 202	- 0 / 0
styles.dart	57.1 % 12 / 21	- 0 / 0
dashboard_page.dart	63.3 % 69 / 109	- 0 / 0
event_functions.dart	70.5 % 31 / 44	- 0 / 0
maintenance_records_options.dart	71.4 % 15 / 21	- 0 / 0
journeys_page.dart	71.5 % 113 / 158	- 0 / 0
editing_popup.dart	71.8 % 183 / 255	- 0 / 0
maintenance_page.dart	72.9 % 191 / 262	- 0 / 0
calendar_page.dart	81.1 % 103 / 127	- 0 / 0
popup_inline_dropdown.dart	83.3 % 30 / 36	- 0 / 0
todolist.dart	88.0 % 73 / 83	- 0 / 0
clients_page.dart	91.7 % 166 / 181	- 0 / 0
main_theme.dart	100.0 % 3 / 3	- 0 / 0
custom_calendar_events.dart	100.0 % 41 / 41	- 0 / 0
event_display.dart	100.0 % 49 / 49	- 0 / 0

Generated by: LCOV version 1.14

Our previous test coverage was stronger, but with the backend overhaul, we had to rewrite all frontend tests to align with the new structure.

Coverage gaps are primarily in non-UI modules e.g., utilities

Early development focused on building features fast – testing lagged in logic layers



Backend Test Coverage

api\models.py	59	0	0	100%
api\serializers.py	64	0	0	100%
api\tests.py	843	0	0	100%
api\urls.py	3	0	0	100%
api\views.py	358	40	0	89%
manage.py	11	2	0	82%
sprintBackend__init__.py	0	0	0	100%
sprintBackend\settings.py	22	0	0	100%
sprintBackend\urls.py	4	0	0	100%
Total	1503	42	0	97%

Django tests written to ensure that each view functions as expected under various conditions



Working Methodology

Issue Creation

The screenshot shows a Kanban board with three columns: **Blocked** (2 items), **ToDo** (14 items), and **In progress** (11 items). Each item is a task card with a title, description, and tags. For example, in the 'Blocked' column, one task is '2024-SprintCommunityTransport #144' with the description '[Custom Template]: Contact Sarah & Peter Oliver about future maintenance of product' and tags 'Priority - Medium!', 'Extra small', 'client interaction', and 'Release'. In the 'ToDo' column, a task is '2024-SprintCommunityTransport #127' with the description '[Feature Request]: Edit client button tests' and tags 'Priority - High!!', 'enhancement', 'features', and 'testing'. In the 'In progress' column, a task is '2024-SprintCommunityTransport #182' with the description '[Feature Request]: Be able to assign a bus to a calendar event' and tags 'Priority - High!!', 'Medium', 'enhancement', and 'features'.

Create new issue

Bug Report

File a bug report.

Custom Template

Create an issue for anything that isn't a bug report or feature request.

Feature Request

Suggest a new feature or improvement for this project.

Blank issue

Create a new issue from scratch

We made three templates for issues (Bug Report, Feature Request & Custom Template) to ensure issues were consistent among team members.

The screenshot shows a Kanban board with two columns: **Testing/Review** (13 items) and **Done** (150 items). Each item is a task card. In the 'Testing/Review' column, a task is '2024-SprintCommunityTransport #138' with the description '[Feature Request]: Login page' and tags 'Priority - High!!', 'Extra large', 'enhancement', and 'features'. In the 'Done' column, a task is '2024-SprintCommunityTransport #146' with the description '[Custom Template]: Set up all team members machines for django backend' and tags 'Priority - High!!', 'Medium', and 'initial setup'.

Using PR template including lots of meta data.

Working Methodology

Detailed PRs with lots of comments and discussion before approval



☐ Django dev ✓ database	#117 by bcart10 was merged on Feb 23 • Approved 9 of 15 tasks Beta Release	10	54
☐ Client journey link ✗ database enhancement features	#221 by bcart10 was merged on Mar 24 • Approved 8 of 15 tasks Final Release	10	39
☐ Improving popup ✓ bug enhancement	#174 by Eva-Anderson was merged on Feb 26 • Approved 5 of 15 tasks Final Release	3	34
☐ Add login page ✓ enhancement features	#118 by Ethanwang-123 was merged on Mar 12 • Approved 7 of 15 tasks Beta Release	1	33
☐ Changes to popup ✓ bug enhancement	#252 by Eva-Anderson was merged on Apr 1 • Approved 9 of 14 tasks Final Release		
☐ Improvements to the maintenance page ✓ enhancement	#183 by sajjad-alkhummais was merged on Mar 6 • Approved 13 of 15 tasks		

Pull Requests

Amylose commented on Feb 5 • edited

Client Meeting Documentation PR

Description

Adding in documentation for our beta release meeting with the client on 07/02/2025. This will include the meeting agenda & minutes. We will then use this information in our Beta Release VIVA.

I have also added some testing documentation into the docs folder! This includes a graph displaying our SUS survey results & a document of key feedback themes

Related Issue

- Closes [Custom Template]: Create Agenda for Beta Release Client Meeting #136
- Closes [Custom Template]: Pull together Testing Day data #133
- Closes [Custom Template]: Meeting minutes from beta handover #143

Type of Change

☐ Bug fix

☐ New feature

☐ Improvement of a current feature

☐ Breaking Change

☒ Documentation

☐ Other (please specify): _____

Reviewers

JamesRenn-1e

bcart10

Assignees

Amylose

Labels

client interaction

documentation

Projects

Sprint Community Transport Main Ka...

Status: Done

+5 more

Milestone

Beta Release

Development

Successfully merging this pull request may close these issues.

[Custom Template]: Meeting minutes from beta ...

[Custom Template]: Create Agenda for Beta Rele...

[Custom Template]: Pull together Testing Day data



Working Methodology



Weekly meetings

Upon reflection after TB1, we moved towards **more agile** style meetings with **weekly standups**

Balancing Workload

Set expectations from **week to week**, giving realistic tasks.
Creation of **roadmaps** to align expectations

Collaboration

Introduced **paired programming** to tackle larger tasks and **rotating project roles**

Communication

GitHub as a **centralised platform** for sharing information.
Coordinating tasks through the **Kanban** board

User Testing

■ Somewhat Agree ■ Agree ■ Neither agree nor disagree ■ Somewhat Disagree ■ Disagree

I needed to learn a lot of things before I could get going with this system.



I felt very confident using the system.



I found the system very cumbersome to use.



I would imagine that most people would learn to use this system very quickly.



I thought there was too much inconsistency in this system.



I found the various functions in this system were well integrated.



I think that I would need the support of a technical person to be able to use this system.



I thought the system was easy to use.



I found the system unnecessarily complex.



I think that I would like to use this system frequently.



Testathon

25 participants

Used SUS
Questionnaire to test
usability

SUS Score:
81.6

Key User Feedback

Agree/Somewhat agree percentages
25 participants

84%

The system was
easy to use

96%

The system would
be easy to learn

80%

Very confident
using the system

Feedback themes



Increasing Input
Validation



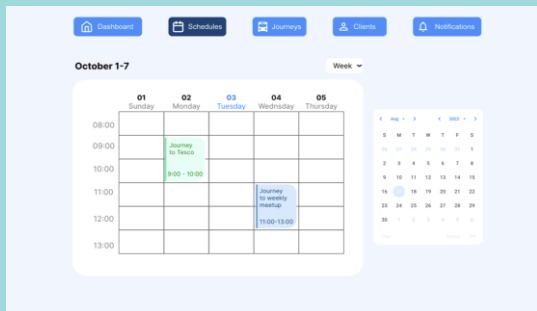
Consistent Design



Calendar Page
functionality

Calendar Page Development

Original UI Design

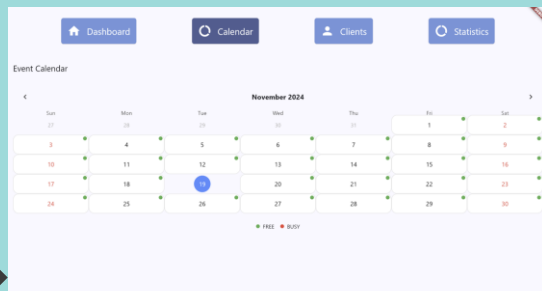


Weekly & monthly views



Clearly visible events

Initial Implementation at MVP



Daily events unclear

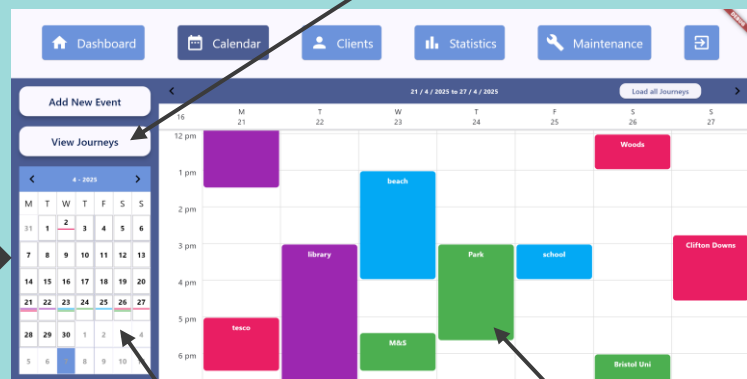


Can't repeat events



No monthly view

Final Implementation



Separate page for repeatable journeys

Monthly view of calendar

Events colour coded by bus

Original UI Design

Initial Implementation at MVP

Final Implementation

Multiple disabilities & special needs per client

Client info
editable

Other columns added:

DOB	Email Address	Postcode	Payment Owed
-----	---------------	----------	--------------

- ✓ Columns for all client info
 - ✓ Colour coded membership status
 - ✓ Table similar to initial design
 - ✗ Clients can only have one disability and special need
 - ✗ Missing columns for necessary information

Client Acceptance Testing Feedback & Solutions

MVP



Maintenance Page

Information required on what upcoming/previous maintenance requirements are for all vehicles



Creation of new
Maintenance Page



Calendar Design

Design of the Calendar/Schedule page should be improved to align more with requirements



New Design &
Implementation

Beta



Schedule Print Outs

Daily schedules need to be provided to each driver with an outline of the day



Functionality
Implemented



Product Maintenance

After our Final Release, will there still be contact?
What happens if something fails?



Extensive Handover
Process



Document tabs



Instruction manual 10

Sprint Community Transp...

Downloading or reinstalli...

Logging In or forgotten p...

Navigating the pages

Dashboard

Daily schedule

ToDo list

Calendar

One off events

Recurring events

Clients

Statistics

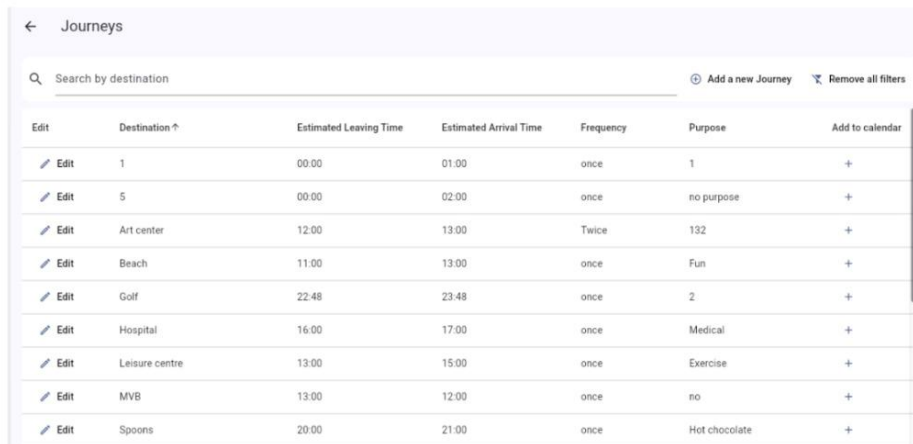
Maintenance

Contact

Recurring events

To add a new recurring journey, click the **"View journeys"** button. This will bring you to a list of all the journey templates that have been created.

A journey template contains all the information about a journey that happens frequently, except for the date.



The screenshot shows the 'Journeys' app interface. At the top, there is a back arrow, the title 'Journeys', a search bar with the placeholder 'Search by destination', and two buttons: '+ Add a new Journey' and 'Remove all filters'. Below the search bar is a table with the following columns: Edit, Destination, Estimated Leaving Time, Estimated Arrival Time, Frequency, Purpose, and Add to calendar. The table contains 10 rows of journey templates.

Edit	Destination	Estimated Leaving Time	Estimated Arrival Time	Frequency	Purpose	Add to calendar
	1	00:00	01:00	once	1	+
	5	00:00	02:00	once	no purpose	+
	Art center	12:00	13:00	Twice	132	+
	Beach	11:00	13:00	once	Fun	+
	Golf	22:48	23:48	once	2	+
	Hospital	16:00	17:00	once	Medical	+
	Leisure centre	13:00	15:00	once	Exercise	+
	MVB	13:00	12:00	once	no	+
	Spoons	20:00	21:00	once	Hot chocolate	+

To create a new journey template, click the **"Add a new journey"** button, which will then take you to a menu allowing you to specify a new journey template.

To search for a specific journey, type the destination in the search bar. To remove all search criteria and view the entire table again, click **"Remove all filters"**

To modify a journey template, click the corresponding **"Edit"** button which will take you to a menu where you can alter the values stored in the template. Then click "

To delete a journey template, click the corresponding **"Edit"** button, and then click the red **"Delete"** button

WARNING: This is permanent and cannot be undone. Only delete if you are sure you will not need this information again.

Final Client Feedback



User Interface Design

Great interface that prioritises usability



When can we have it?!

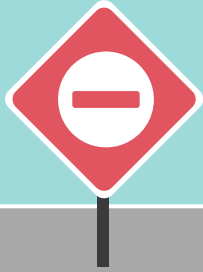
Excited to implement into their workflow



Streamlined Onboarding

Easy to teach new team members

Challenges Encountered



**Requirements
gathering with
an untechnical
client**

- Avoiding technical language
- Providing options
- Frequent check-ins

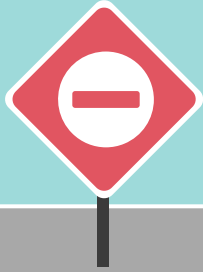
Challenges Encountered



Project's Scope Shifting

- Roadmaps for required features
- Hard deadlines for fundamentals
- Focus on main User Stories

Challenges Encountered



**Teammate
Injuries : Amy's
Shoulder**

- Switching client liason role
- Taking on more research tasks
- Team Communication

SEP in Review: What would we do differently?



Requirements Gathering

Clearer initial communication could have prevented later changes.



Agile Workflow

Working in an agile framework from the start, setting more precise goals each week



Standardisation

Make use of style sheets & code formatters for more uniform code based



Test driven development

Putting testing at the forefront of development

Questions?

